

MODULAR ARITHMETIC OF RAPIDLY GROWING POWERS

Delayed digit stabilization**A counterexample, exact minima, and sharp bounds**

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Abstract

For $a \geq 2$ not divisible by 10, put $S_a(b) = v_{10}(a^{10^{b+1}} - a^{10^b})$ and $D_a(b) = S_a(b) - S_a(b-1)$. A publicly posed conjecture asks whether $D_a(b) = 1$ always holds for $b \geq 3$. We disprove it and prove that the smallest counterexample is $a_\star = 3 \cdot 2^{99} + 1 = 1901475900342344102245054808065$: $S_{a_\star}(2) = 100$ and $S_{a_\star}(3) = 102$. The mechanism is competition between an exponentially growing valuation at one prime and an affine valuation at the other. We give exact formulas for all bases, determine the first permanently unit increment, construct arbitrarily delayed examples, and determine the least counterexample at every prescribed stage. An exact density formula quantifies the scarcity of exceptions. We then prove an arbitrary-radix local formula, characterize convergence and finite limit cycles, and explain how fast-growing changes of index transform the stabilization rate. A separate calculation for genuine integer tetration makes the distinction between the two iterations explicit. All mathematical claims are proved in ordinary mathematics; accompanying standard-library Python code supplies independent modular certificates and regression tests.

Outcome. The uniform threshold is false, not merely unproved. Eventual unit speed for each fixed admissible base is true. The exact threshold is unbounded as the base varies, even though it grows only on a $\log \log a$ scale in the worst case.

Keywords. Trailing digits; lifting the exponent; tetration; p -adic valuation; finite iteration; Chinese remainder theorem.

Scope. The conjecture concerns a^{10^b} , not $a \uparrow\uparrow b$. The relation to tetration and higher-growth indexing is developed in Sections 9 and 12.

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1 The selected problem and its status

1.1 A concrete, independently checkable target

The target is a question posed by Marco Ripà on Mathematics Stack Exchange on 24 October 2025 [1]. With the notation used in the abstract, it asks whether every integer $a > 1$ with $10 \nmid a$ satisfies

$$D_a(b) = 1 \quad \text{for every } b \geq 3. \quad (1)$$

The source connects this question to earlier work on the congruence speed of integer tetration. Its retrieved public page displayed no answers during the literature check for this note. A targeted search did not locate a resolution of this exact question. That is a record of the checked sources, not a claim that every possible prior publication has been excluded.

The advantage of this target is that the conjectural assertion has precise quantifiers and admits a finite certificate of failure. There is no need to choose an analytic extension of tetration or a convention for ordinal fundamental sequences. More importantly, the question can be reduced to a competition between two exactly computable integer quantities.

1.2 What is proved, and what is inherited

The proof of the counterexample is in Section 4; its minimality is also proved there. The stronger least-counterexample theorem for every stage appears in Section 6. The density calculation and the arbitrary-radix results are additional deductions developed here. Their proofs, rather than an assertion of novelty, are the basis for the claims made in this note.

The lifting identities used below are classical. The question already reports a base-dependent sufficient onset bound [1]; pointwise eventual unit speed is not claimed here as new. Earlier tetration papers establish valuation formulas and eventual constant congruence speeds for integer towers [2, 3]. Those results are relevant background, but they must not be substituted for the formulas of this note: the exponent schedule is different. Work on the stable digits of Graham's number provides another explicit application of valuation methods to very large integers [4]. We do not claim a new solution of that problem.

All calculations here use positive integer bases. Bases divisible by 10 are included in the general formulas but excluded from the conjecture. The base 1 is excluded because every difference is zero and its valuation would be infinite.

1.3 The mathematical mechanism in one paragraph

An even base can be exceptionally close to a fourth root of unity in the 5-adic metric while being divisible by exactly one power of 2. Conversely, an odd multiple of 5 can be exceptionally close to 1 or -1 in the 2-adic metric while containing only one factor of 5. The number of matching decimal digits is the *smaller* of the two local valuations. A large constant in the affine local valuation can therefore postpone the point at which it becomes the smaller one. This is the missing uniformity in (1).

2 Valuations and the lifting identities

For a prime p and a nonzero integer z , let $v_p(z)$ be the largest integer $k \geq 0$ such that $p^k \mid z$. For any radix $B \geq 2$, define

$$v_B(z) = \max\{k \geq 0 : B^k \mid z\}. \quad (2)$$

If $B = \prod_{p|B} p^{e_p}$, then

$$v_B(z) = \min_{p|B} \left\lfloor \frac{v_p(z)}{e_p} \right\rfloor. \quad (3)$$

This follows by comparing the prime exponents in B^k and z . In particular, $v_{10}(z) = \min\{v_2(z), v_5(z)\}$. For a composite radix this is a divisibility count; it is not generally an additive prime valuation.

Two nonnegative integers agree in their last k base- B positions exactly when their difference is divisible by B^k . Leading zero padding is understood when necessary.

Lemma 2.1 (Odd-prime lifting). *Let p be odd and let $p \mid u - 1$. For every positive integer m ,*

$$v_p(u^m - 1) = v_p(u - 1) + v_p(m).$$

More generally, if $p \nmid a$ and $d = \text{ord}_p(a)$, then

$$v_p(a^m - 1) = \begin{cases} 0, & d \nmid m, \\ v_p(a^d - 1) + v_p(m), & d \mid m. \end{cases}$$

Proof. Write $m = p^j w$, where $p \nmid w$. If $u \equiv 1 \pmod{p}$, then $(u^w - 1)/(u - 1)$ is congruent to w modulo p , so raising to the power w does not change the valuation. If $u = 1 + t$ with $v_p(t) = r \geq 1$, the first term of

$$(1 + t)^p - 1 = pt + \binom{p}{2}t^2 + \cdots + t^p$$

has valuation $r + 1$. Every other term has valuation at least $r + 2$. Thus each subsequent raising to the power p increases the valuation by exactly one. This proves the first identity.

For the second statement, $a^m \equiv 1 \pmod{p}$ holds exactly when $d \mid m$. In that case apply the first identity to $u = a^d$ and exponent m/d . Since $d \mid p - 1$, we have $p \nmid d$ and $v_p(m/d) = v_p(m)$. $\square$

Lemma 2.2 (Binary lifting). *For odd a and positive even m ,*

$$v_2(a^m - 1) = v_2(a - 1) + v_2(a + 1) + v_2(m) - 1. \quad (4)$$

Proof. Write $m = 2^j w$ with $j \geq 1$ and w odd. Odd powers leave the valuation of $u - 1$ unchanged, since $(u^w - 1)/(u - 1)$ is an odd sum of odd terms. The first squaring contributes $v_2(a^2 - 1) = v_2(a - 1) + v_2(a + 1)$, which is at least 3. After this squaring the current value is 1 modulo 8, so each further factor $u + 1$ contributes exactly one factor of 2. There are $j - 1$ such further squarings, giving (4). $\square$

Lemma 2.3 (A prime dividing the base). *If $p \mid a$ and $0 < u < v$, then*

$$v_p(a^v - a^u) = uv_p(a).$$

Proof. Factor the difference as $a^u(a^{v-u} - 1)$. The second factor is -1 modulo p and contributes no further factors of p . $\square$

3 The complete decimal profile

For an odd integer $a > 1$, put

$$C_2(a) = v_2(a^2 - 1) - 1 = \max\{v_2(a - 1), v_2(a + 1)\}. \quad (5)$$

The equality holds because exactly one of $a - 1, a + 1$ is divisible by 4, and the other has valuation 1. For $5 \nmid a$, put

$$C_5(a) = v_5(a^4 - 1). \quad (6)$$

The factorization of $a^4 - 1$ gives the useful equivalent formula

$$C_5(a) = \begin{cases} v_5(a-1), & a \equiv 1 \pmod{5}, \\ v_5(a+1), & a \equiv 4 \pmod{5}, \\ v_5(a^2+1), & a \equiv 2, 3 \pmod{5}. \end{cases} \quad (7)$$

Only one of the displayed factors is divisible by 5.

Theorem 3.1 (Exact profile for every decimal base). *Let $a \geq 2$ and $b \geq 2$. Then*

$$S_a(b) = \begin{cases} b + \min\{C_2(a), C_5(a)\}, & \gcd(a, 10) = 1, \\ \min\{v_2(a)10^b, b + C_5(a)\}, & 2 \mid a, 5 \nmid a, \\ \min\{v_5(a)10^b, b + C_2(a)\}, & 5 \mid a, 2 \nmid a, \\ 10^b \min\{v_2(a), v_5(a)\}, & 10 \mid a. \end{cases} \quad (8)$$

Proof. The fundamental factorization is

$$a^{10^{b+1}} - a^{10^b} = a^{10^b} (a^{9 \cdot 10^b} - 1). \quad (9)$$

If $2 \mid a$, Lemma 2.3 gives valuation $v_2(a)10^b$. If a is odd, Lemma 2.2 applies with $v_2(9 \cdot 10^b) = b$ and gives $b + C_2(a)$.

If $5 \mid a$, Lemma 2.3 gives valuation $v_5(a)10^b$. If $5 \nmid a$, then $\text{ord}_5(a) \mid 4$, while $4 \mid 10^b$ because $b \geq 2$. Apply odd-prime lifting to a^4 with exponent $9 \cdot 10^b/4$. The result is $b + C_5(a)$. Taking the smaller of the binary and quinary valuations proves all four cases. $\square$

The hypothesis $b \geq 2$ is substantive: when $b = 1$, the exponent $9 \cdot 10 = 90$ need not be divisible by 4. The conjectured increments start at $b = 3$, so both $S_a(b)$ and $S_a(b-1)$ lie in the range where the theorem applies.

Corollary 3.2 (The two different assertions). *For every fixed $a \geq 2$ with $10 \nmid a$, there is an integer $C(a)$ such that $S_a(b) = b + C(a)$ for all sufficiently large b . If $\gcd(a, 10) = 1$, this already holds for every $b \geq 2$.*

Proof. The coprime case is explicit in (8). In either mixed case, $t10^b > b + C$ eventually, for the fixed positive integers t and C . $\square$

This proves pointwise eventual stabilization. It gives no base-independent bound on the initial exceptional interval.

4 A counterexample and the proof of its minimality

4.1 The counterexample

Theorem 4.1. *The integer*

$$a_\star = 3 \cdot 2^{99} + 1 = 1901475900342344102245054808065 \quad (10)$$

is a counterexample to (1). Specifically,

$$S_{a_\star}(2) = 100, \quad S_{a_\star}(3) = 102, \quad D_{a_\star}(3) = 2.$$

Proof. We have $2^{99} \equiv 13 \pmod{25}$, so $a_\star \equiv 15 \pmod{25}$. Consequently $v_5(a_\star) = 1$. Moreover,

$$v_2(a_\star - 1) = 99, \quad v_2(a_\star + 1) = 1,$$

and hence $C_2(a_\star) = 99$. The odd-multiple-of-5 case of Theorem 3.1 now gives

$$S_{a_\star}(b) = \min\{10^b, b + 99\} \quad (b \geq 2). \quad (11)$$

At $b = 2$ the minimum is 100, whereas at $b = 3$ it is 102. □

b	$v_2(\Delta_b)$	$v_5(\Delta_b)$	$S_{a_\star}(b)$	$D_{a_\star}(b)$
2	101	100	100	—
3	102	1000	102	2
4	103	10000	103	1
5	104	100000	104	1
6	105	1000000	105	1

$$\text{Here } \Delta_b = a_\star^{10^{b+1}} - a_\star^{10^b}.$$

The exceptional increment is caused by a change in which prime controls the minimum. No approximation of the enormous powers is involved.

4.2 An exact test for unit increments

Lemma 4.2. *Let t, C be positive integers, and let $S(b) = \min\{t10^b, b + C\}$ for $b \geq 2$. For $b \geq 3$,*

$$S(b) - S(b-1) = 1 \iff C \leq t10^{b-1} - (b-1). \quad (12)$$

If the inequality fails, the increment is at least 2.

Proof. Write $R = t10^{b-1}$ and $L = b - 1 + C$. If $L \leq R$, then the previous minimum is L and the next is $L + 1$, because $L + 1 \leq R + 1 \leq 10R$. If $L > R$, integrality gives $L \geq R + 1$. The previous minimum is R and the next is $\min\{10R, L + 1\} \geq R + 2$. □

Since $t10^{b-1} - (b-1)$ strictly increases with b , once a unit increment occurs, every later increment is also one. In particular, a base satisfies the conjecture at every stage $b \geq 3$ exactly when it satisfies it at $b = 3$.

4.3 Why no smaller integer works

Theorem 4.3 (Global minimality). *The integer $a_\star$ of (10) is the smallest $a \geq 2$, $10 \nmid a$, for which $D_a(b) \neq 1$ at any stage $b \geq 3$.*

Proof. It suffices to rule out $D_a(3) \neq 1$ for $a < a_\star$. Coprime bases cannot fail, by Theorem 3.1.

Suppose first that a is even and $5 \nmid a$. By Lemma 4.2, failure requires

$$C_5(a) \geq 100v_2(a) - 1 \geq 99.$$

By (7), one of $a - 1, a + 1, a^2 + 1$ is divisible by 5^{99} . Thus $a^2 + 1 \geq 5^{99}$. But

$$a_\star^2 + 1 < 5^{99}, \quad (13)$$

so this is impossible when $a < a_*$. For an elementary check of the bound, $a_* < 4 \cdot 2^{99}$, and $16 \cdot 4^{99} + 1 < 17 \cdot 4^{99} < 5^{99}$. The latter inequality follows already from $(5/4)^{16} > 25 > 17$ and $99 > 16$.

Now suppose that a is odd and $5 \mid a$. Failure requires

$$C_2(a) \geq 100v_5(a) - 1 \geq 99.$$

Hence $a \equiv 1$ or $-1 \pmod{2^{99}}$. Among the integers between 2 and $3 \cdot 2^{99} + 1$, the only possible smaller multiple of 5 in these two residue classes is $2^{100} - 1$. This can be checked by considering $j2^{99} \pm 1$ for $j = 1, 2, 3$ and using $2^{99} \equiv 3 \pmod{5}$. For that remaining candidate,

$$C_2(2^{100} - 1) = 100, \quad v_5(2^{100} - 1) = v_5(16^{25} - 1) = 1 + 2 = 3.$$

Its failure condition would be $100 \geq 299$, which is false. Thus every smaller admissible base satisfies the conjectured bound. $\square$

The size of the first counterexample explains why a search restricted to small bases offers misleading evidence for the uniform statement. This is a conclusion from a minimality proof, not an extrapolation of an experiment.

5 The exact onset and arbitrarily long delays

Define the onset, restricted to the range relevant to the conjecture, by

$$N(a) = \min\{B \geq 3 : D_a(b) = 1 \text{ for all } b \geq B\}. \quad (14)$$

It exists for every admissible a by Corollary 3.2.

Theorem 5.1 (Exact onset). *If $\gcd(a, 10) = 1$, then $N(a) = 3$. In either mixed case, use*

$$(t, C) = \begin{cases} (v_2(a), C_5(a)), & 2 \mid a, 5 \nmid a, \\ (v_5(a), C_2(a)), & 5 \mid a, 2 \nmid a. \end{cases}$$

Then

$$N(a) = 1 + \min\{r \geq 2 : t10^r - r \geq C\}. \quad (15)$$

Proof. Substitute $r = b - 1$ in Lemma 4.2; the inequality is monotone in r . $\square$

This is a sharp characterization, not merely a sufficient bound. If

$$r_0 = \max\{2, \lceil \log_{10}(C/t) \rceil\},$$

the minimum in (15) is either r_0 or $r_0 + 1$. Indeed, smaller admissible r cannot make $t10^r \geq C$, and the additional factor of ten at the next stage more than compensates for subtracting r . The exact formula can be evaluated by integer comparisons without computing any logarithms.

Theorem 5.2 (A family realizing every later onset). *For $m \geq 2$, set*

$$K_m = 10^m - m + 1, \quad a_m = 5^{K_m} + 1. \quad (16)$$

Then

$$D_{a_m}(m+1) = 2, \quad N(a_m) = m+2.$$

In particular, no absolute onset bound is valid for all admissible bases.

Proof. For every positive integer K , the number $5^K + 1$ is 2 modulo 4 and 1 modulo 5. Thus $v_2(5^K + 1) = 1$ and $C_5(5^K + 1) = K$, so

$$S_{5^K+1}(b) = \min\{10^b, b + K\}.$$

At $K = K_m$, the linear term exceeds 10^m by one when $b = m$. Hence $S_{a_m}(m) = 10^m$, while $S_{a_m}(m+1) = 10^m + 2$. The criterion fails at $r = m$ and holds at $r = m+1$; because it is monotone, the asserted onset is exact. $\square$

The family $5^K + 1$ gives $N(5^K + 1) = \log_{10} K + O(1)$ as $K \rightarrow \infty$. More generally,

$$N(a) = O(\log \log a) \tag{17}$$

as a ranges over admissible bases. To see this, use $C_2(a) \leq \log_2(a+1)$ and $C_5(a) \leq \log_5(a^2+1)$ in the onset formula, together with $t \geq 1$. The family above also shows that the worst-case order in (17) cannot be improved to $o(\log \log a)$. Unboundedness and slow growth are entirely compatible.

6 The least counterexample at every stage

The preceding family is convenient but not usually minimal. We can solve the minimization problem exactly.

Definition 6.1. For $b \geq 3$, let

$$A_b = \min\{a \geq 2 : 10 \nmid a, D_a(b) \neq 1\}, \quad K = 10^{b-1} - b + 2.$$

The set is nonempty by Theorem 5.2.

Theorem 6.2 (Exact stage minima). *For $b \geq 3$, the least exceptional base is the following integer:*

Condition on K	A_b	$D_{A_b}(b)$
$K \equiv 0 \pmod{4}, K \not\equiv 0 \pmod{20}$	$2^K - 1$	2
$K \equiv 0 \pmod{20}$	$4 \cdot 2^K + 1$	4
$K \equiv 1 \pmod{4}, K \not\equiv 9 \pmod{20}$	$2 \cdot 2^K + 1$	3
$K \equiv 9 \pmod{20}$	$3 \cdot 2^K - 1$	2
$K \equiv 2 \pmod{4}, K \not\equiv 10 \pmod{20}$	$2^K + 1$	2
$K \equiv 10 \pmod{20}$	$4 \cdot 2^K - 1$	4
$K \equiv 3 \pmod{4}, K \not\equiv 19 \pmod{20}$	$2 \cdot 2^K - 1$	3
$K \equiv 19 \pmod{20}$	$3 \cdot 2^K + 1$	2

In every case $N(A_b) = b + 1$.

Proof. Put $r = b - 1$ and $Q = 10^r$, so $K = Q - r + 1 \geq 99$. For a mixed base, the failure condition is

$$C \geq tQ - r + 1. \tag{18}$$

We will find the minimum below $4 \cdot 2^K + 1$.

An even exceptional base must have $C_5(a) \geq K$, so $a^2 + 1 \geq 5^K$. But for $K \geq 99$,

$$(4 \cdot 2^K + 1)^2 + 1 < 25 \cdot 4^K < 5^K.$$

The first inequality holds already for $K \geq 1$; the second follows from $(5/4)^{16} > 25$ and $K \geq 99$. No even base in the proposed range can therefore be exceptional.

For an odd multiple of 5 in this range, $C_2(a) \leq K + 2$. If $v_5(a) \geq 2$, condition (18) requires

$$C_2(a) \geq 2Q - r + 1 = 2K + r - 1 > K + 2,$$

which is impossible. Thus we need precisely

$$v_5(a) = 1, \quad a \equiv \pm 1 \pmod{2^K}. \quad (19)$$

Every candidate in the range, other than the excluded integer 1, is $j2^K \pm 1$ for $1 \leq j \leq 4$.

The residue of 2^K modulo 5 gives the first possible multiple of 5. For K congruent to 0, 1, 2, 3 modulo 4, respectively, these first candidates are

$$2^K - 1, \quad 2 \cdot 2^K + 1, \quad 2^K + 1, \quad 2 \cdot 2^K - 1.$$

A candidate must be skipped exactly when it is also divisible by 25. The order of 2 modulo 25 is 20, with $2^{10} \equiv -1 \pmod{25}$. The four exceptional residue classes are consequently $K \equiv 0, 9, 10, 19 \pmod{20}$, respectively. The next candidates are $4 \cdot 2^K + 1$, $3 \cdot 2^K - 1$, $4 \cdot 2^K - 1$, and $3 \cdot 2^K + 1$. Direct reduction modulo 25 shows that each has exactly one factor of 5. This proves the table and its minimality.

If the selected candidate is $A_b = j2^K \pm 1$, then $C_2(A_b) = K + v_2(j)$. At stage $b - 1$, its profile is Q ; at stage b , it is $b + K + v_2(j)$. Their difference is $2 + v_2(j)$, giving the last column. At the next stage the inequality of Lemma 4.2 holds, so $N(A_b) = b + 1$. $\square$

Because $10^{b-1} \equiv 0 \pmod{20}$, the coefficient and sign in the table depend only on b modulo 20: indeed $K \equiv 2 - b \pmod{20}$. The first few values are

b	K	A_b
3	99	$3 \cdot 2^{99} + 1$
4	998	$2^{998} + 1$
5	9997	$2^{9998} + 1$
6	99996	$2^{99996} - 1$
7	999995	$2^{999996} - 1$

Corollary 6.3 (An exact extremal description). *For $X \geq 2$, let $M(X)$ be the maximum of $N(a)$ over admissible integers $a \leq X$. If $X < A_3$, then $M(X) = 3$. Otherwise,*

$$M(X) = 1 + \max\{b \geq 3 : A_b \leq X\}.$$

In particular,

$$\log_2 A_b = 10^{b-1} - b + O(1), \quad M(X) = \log_{10} \log_2 X + O(1) \quad (X \rightarrow \infty).$$

Proof. The statement $N(a) > b$ is equivalent to $D_a(b) \neq 1$. The definition of A_b therefore characterizes exactly when the set $a \leq X$ contains a base with onset exceeding b . The asymptotic estimates follow from $A_b = j2^K \pm 1$ with $1 \leq j \leq 4$. $\square$

7 How sparse are the counterexamples?

For a fixed $b \geq 3$, write

$$E_b = \{a \geq 2 : 10 \nmid a, D_a(b) \neq 1\}.$$

We compute its natural density among *all* positive integers:

$$\delta_b = \lim_{X \rightarrow \infty} \frac{\#(E_b \cap [1, X])}{X}.$$

The conditional density among admissible bases is $10\delta_b/9$.

Theorem 7.1 (Exact exceptional density). *Let $r = b - 1$ and $Q = 10^r$. The density exists and equals*

$$\delta_b = \frac{2 \cdot 5^{r-1}}{2 \cdot 5Q - 1} + \frac{2^{r+2}}{5(5 \cdot 2Q - 1)}. \quad (20)$$

The first term counts even bases; the second counts odd multiples of 5.

Proof. Fix an even base with $v_2(a) = t \geq 1$. Failure means

$$v_5(a^4 - 1) \geq K_t, \quad K_t = tQ - r + 1.$$

There are exactly four roots of $x^4 = 1$ modulo 5^{K_t} . Indeed, modulo 5 all four nonzero classes are roots, and the derivative $4x^3$ is nonzero at each root. To see the unique lifting directly, replace a root x modulo 5^k by $x + j5^k$; the coefficient of j in the next congruence is invertible modulo 5, determining exactly one j . The density of the quinary condition is thus $4/5^{K_t}$. The exact binary valuation condition has density $1/2^{t+1}$. By the Chinese remainder theorem their intersection has density

$$\frac{4}{2^{t+1}5^{tQ-r+1}}.$$

Summing over $t \geq 1$ gives

$$2 \cdot 5^{r-1} \sum_{t \geq 1} \left(\frac{1}{2 \cdot 5Q} \right)^t = \frac{2 \cdot 5^{r-1}}{2 \cdot 5Q - 1}.$$

For odd multiples of 5 with $v_5(a) = s \geq 1$, the density of the exact quinary condition is $4/5^{s+1}$. Failure requires

$$C_2(a) \geq K_s = sQ - r + 1.$$

Since $K_s \geq 99$, this is exactly $a \equiv 1$ or $-1 \pmod{2^{K_s}}$, a condition of density $2/2^{K_s}$. Again applying the Chinese remainder theorem gives density

$$\frac{8}{5^{s+1}2^{sQ-r+1}}.$$

Its sum over $s \geq 1$ is

$$\frac{2^{r+2}}{5} \sum_{s \geq 1} \left(\frac{1}{5 \cdot 2Q} \right)^s = \frac{2^{r+2}}{5(5 \cdot 2Q - 1)}.$$

It remains to justify summing infinitely many densities. After truncating the even-base sum at $t < T$, its omitted set is contained in the multiples of 2^T , whose upper density is 2^{-T} . The corresponding omitted odd-base set is contained in the multiples of 5^T , of upper density 5^{-T} . These bounds tend to zero. Finite sums have densities by the Chinese remainder theorem, so the infinite union has the stated density. $\square$

For the original conjecture,

$$\delta_3 = \frac{10}{2 \cdot 5^{100} - 1} + \frac{16}{5(5 \cdot 2^{100} - 1)} \approx 5.0487097934144756 \times 10^{-31}. \quad (21)$$

This is a natural density, not an independence assumption about random samples. In particular, it should not be interpreted as a theorem about the distribution of exceptions in every finite interval.

The second term dominates as $b \rightarrow \infty$, so

$$\delta_b \sim \frac{2^{b+1-10^{b-1}}}{25}. \quad (22)$$

The extraordinary rarity of exceptions is therefore quantified by a closed formula, while their existence at every stage is guaranteed by the explicit constructions.

8 Permanent digits and the four decimal limits

So far $S_a(b)$ was defined using neighboring terms. The same number actually measures agreement with *every* later term.

Theorem 8.1 (Exact nonconsecutive agreement). *For every $a \geq 2$ and integers $v > u \geq 2$,*

$$v_{10}(a^{10^v} - a^{10^u}) = S_a(u). \quad (23)$$

Proof. Factor the difference as

$$a^{10^u} (a^{10^u(10^{v-u}-1)} - 1).$$

The number $10^{v-u} - 1$ is coprime to 10. Consequently the exponent in the second factor has binary and quinary valuations both equal to u , and is divisible by 4. Every step of the proof of Theorem 3.1 therefore gives the same local valuations as for the neighboring-term difference. $\square$

The 10-adic integers may be defined as compatible systems of residues modulo 10^k . The Chinese remainder theorem gives

$$\mathbb{Z}_{10} \cong \mathbb{Z}_2 \times \mathbb{Z}_5.$$

This is a product of two rings, not a field. Since $S_a(u) \rightarrow \infty$, Theorem 8.1 shows that a^{10^u} has a limit $L_a \in \mathbb{Z}_{10}$.

Theorem 8.2 (Limit and exact distance). *The limit depends only on which of 2 and 5 divide a :*

$$L_a \longleftrightarrow (\mathbf{1}_{2 \nmid a}, \mathbf{1}_{5 \nmid a}) \quad \text{in } \mathbb{Z}_2 \times \mathbb{Z}_5. \quad (24)$$

These are the four idempotents of $\mathbb{Z}_{10}$. Moreover, for $u \geq 2$,

$$v_{10}(L_a - a^{10^u}) = S_a(u). \quad (25)$$

Proof. If $p \mid a$, then $v_p(a^{10^u}) = 10^u v_p(a) \rightarrow \infty$, so the p -adic component tends to zero. If a is odd, binary lifting gives $v_2(a^{10^u} - 1) = u + C_2(a) \rightarrow \infty$. If $5 \nmid a$ and $u \geq 2$, odd-prime lifting gives $v_5(a^{10^u} - 1) = u + C_5(a) \rightarrow \infty$. This proves (24). In each $\mathbb{Z}_p$, the equation $x(x-1) = 0$ has only the solutions 0, 1, since $\mathbb{Z}_p$ has no zero divisors. Thus the product ring has exactly the four displayed idempotents.

For the exact distance, $S_a(u+1) > S_a(u)$ in all cases of Theorem 3.1. Every later difference from $a^{10^{u+1}}$ is divisible by $10^{S_a(u)+1}$. Passing to the limit gives

$$L_a - a^{10^u} \equiv a^{10^{u+1}} - a^{10^u} \pmod{10^{S_a(u)+1}}.$$

The right side has valuation exactly $S_a(u)$, proving (25). $\square$

The finite residue of L_a can be computed without iteration. To find its last k digits, solve

$$z \equiv \mathbf{1}_{2 \nmid a} \pmod{2^k}, \quad z \equiv \mathbf{1}_{5 \nmid a} \pmod{5^k}$$

by the Chinese remainder theorem. Thus the asymptotic limit is elementary; the interesting issue is the exact rate of approach and its nonuniform initial behavior.

9 Fast-growing changes of index

Let $g : \mathbb{N} \rightarrow \mathbb{N}$ be any strictly increasing function with $g(n) \geq 2$, and consider the sampled sequence

$$Y_n = a^{10^{g(n)}}.$$

No computability or growth-rate assumption is needed for the next statement.

Theorem 9.1 (A time-change identity). *For every n ,*

$$\nu_{10}(Y_{n+1} - Y_n) = S_a(g(n)).$$

For fixed $a \geq 2$ with $10 \nmid a$, there is a fixed integer $C(a)$ such that eventually

$$\nu_{10}(Y_{n+1} - Y_n) = g(n) + C(a), \quad (26)$$

$$\nu_{10}(Y_{n+1} - Y_n) - \nu_{10}(Y_n - Y_{n-1}) = g(n) - g(n-1). \quad (27)$$

Proof. Apply Theorem 8.1 with $u = g(n)$ and $v = g(n+1)$. A strictly increasing integer-valued g is unbounded, so eventually $g(n)$ lies in the affine part of the fixed-base profile. Subtract the two consecutive affine formulas. $\square$

For example, choose $g(n) = 2 \uparrow\uparrow n$ for $n \geq 2$, where $2 \uparrow\uparrow n$ denotes the right-associated tower of n copies of 2. Then the exact eventual count of new stable digits per step is

$$2 \uparrow\uparrow n - 2 \uparrow\uparrow (n-1).$$

The same identity applies to any specified increasing integer-valued schedule obtained from higher up-arrow expressions, a chain notation, or a chosen fast-growing hierarchy. The theorem concerns sampling an already defined sequence; it does not assert identities between those different large-number notations.

This distinction matters conceptually. A stabilization rate is attached not just to the limit and base but also to the chosen indexing scheme. Passing from n to a tetration schedule can turn unit increments into tetrational large increments without changing the limiting 10-adic integer.

10 An exact local formula in every radix

Now let $B \geq 2$ be arbitrary, with $B = \prod_{p|B} p^{e_p}$. For $a \geq 2$, $n \geq 1$, and a positive lag h , put

$$S_{a,B,h}(n) = \nu_B(a^{B^{n+h}} - a^{B^n}).$$

When $h = 1$ write $S_{a,B}(n)$.

For an odd prime $p \mid B$ with $p \nmid a$, define

$$d_p = \text{ord}_p(a), \quad c_p = \nu_p(a^{d_p} - 1).$$

For $2 \mid B$ and odd a , put $c_2 = \nu_2(a^2 - 1) - 1$.

Theorem 10.1 (Arbitrary-radix profile). *For every $n, h \geq 1$,*

$$S_{a,B,h}(n) = \min_{p|B} \left\lfloor \frac{w_p(n, h)}{e_p} \right\rfloor, \quad (28)$$

where

$$w_p(n, h) = \begin{cases} \nu_p(a)B^n, & p \mid a, \\ ne_2 + c_2, & p = 2, 2 \nmid a, \\ 0, & p > 2, p \nmid a, d_p \nmid B^n(B^h - 1), \\ ne_p + c_p, & p > 2, p \nmid a, d_p \mid B^n(B^h - 1). \end{cases} \quad (29)$$

Proof. Factor the difference as

$$a^{B^n} \left(a^{B^n(B^h-1)} - 1 \right).$$

For $p \mid a$, use Lemma 2.3. For a unit at an odd p , use Lemma 2.1. Since $p \mid B$, the factor $B^h - 1$ is coprime to p , and the exponent has valuation ne_p . At $p = 2$, the exponent is even and the same observation applies to Lemma 2.2. Finally use (3). $\square$

The multiplicative-order condition cannot be omitted. It is automatically satisfied for decimal bases once $n \geq 2$, but not for arbitrary radices. The floors cannot be omitted either when B contains prime powers.

10.1 Convergence and finite limit cycles

For an odd unit prime $p \mid B$, let $d_p^\perp$ be the largest divisor of d_p coprime to B : remove from d_p all prime factors that divide B .

Theorem 10.2 (Convergence criterion and cycle length). *The sequence a^{B^n} converges in $\mathbb{Z}_B$ if and only if*

$$d_p^\perp \mid B - 1 \quad \text{for every odd } p \mid B \text{ with } p \nmid a. \quad (30)$$

In general it has an asymptotic cycle whose exact length is

$$H = \text{lcm}_{p \mid B, \substack{p > 2 \\ p \nmid a}} \text{ord}_{d_p^\perp}(B), \quad (31)$$

where an order modulo 1, and an empty least common multiple, are taken to be 1. Precisely, each sequence with indices in one residue class modulo H converges, and H is the least positive integer with that property. If $H = 1$, the limit L satisfies $L^B = L$.

Proof. For sufficiently large n , every prime-power divisor of d_p supported on the primes dividing B divides B^n . The remaining factor $d_p^\perp$ is coprime to B^n . Thus, eventually,

$$d_p \mid B^n(B^h - 1) \iff d_p^\perp \mid B^h - 1.$$

For $h = 1$, if one condition fails, adjacent terms remain different modulo that prime p , so the sequence cannot converge. If every condition holds, Theorem 10.1 makes every local valuation tend to infinity. For each fixed precision all sufficiently late adjacent differences vanish; then any finite sum of such differences vanishes at that precision. The sequence is Cauchy and hence converges in the inverse-limit ring $\mathbb{Z}_B$.

The smallest positive h satisfying all the displayed divisibilities is exactly (31). For that lag, the same valuation argument makes every residue-class subsequence Cauchy. If a smaller positive lag worked, its adjacent subsequence terms would eventually agree modulo every $p \mid B$, forcing all the divisibilities and contradicting the minimality of H . Finally $X_{n+1} = X_n^B$ and continuity of polynomial maps show that the limiting points are carried cyclically to one another; in the convergent case this gives $L^B = L$. $\square$

Example 10.3 (An obstruction in radix 14). Take $a = 2$, $B = 14$. Modulo 7, the order of 2 is 3, and $3 \nmid 14 - 1$. Thus $S_{2,14}(n) = 0$ for every $n \geq 1$; adjacent terms never share even one trailing radix-14 digit. Since $14 \equiv -1 \pmod{3}$, the cycle length is 2. For lag 2 the local formulas give

$$S_{2,14,2}(n) = \min\{14^n, n + 1\} = n + 1 \quad (n \geq 1).$$

The even and odd index subsequences therefore converge separately with an exact affine agreement count.

10.2 The eventual affine profile

Assume (30), and suppose at least one prime divisor of B does not divide a . Once the order conditions have become satisfied, Theorem 10.1 reduces to

$$S_{a,B}(n) = \min \left\{ n + C_B(a), \min_{\substack{p|B \\ p \nmid a}} \left\lfloor \frac{\nu_p(a)B^n}{e_p} \right\rfloor \right\}, \quad (32)$$

where

$$C_B(a) = \min_{\substack{p|B \\ p \nmid a}} \left\lfloor \frac{c_p}{e_p} \right\rfloor.$$

The inner minimum over an empty set is omitted. Thus the eventual slope is one whenever there is a unit component and convergence occurs. If every prime dividing B divides a , all components instead have exponential profiles and the limit is zero.

For nonsquarefree B , the condition $B \nmid a$ does *not* guarantee a unit component. For instance $12 \nmid 6$, but both prime divisors of 12 divide 6. The correct condition is $\text{rad}(B) \nmid a$.

11 When can a radix have a uniform onset?

To compare radices fairly, restrict to bases for which a^{B^n} converges and at least one prime divisor of B is a unit prime for a . These are the bases for which the eventual unit-slope conclusion applies.

Theorem 11.1 (Prime powers versus mixed radices). *If B is a prime power, every base in this class has $S_{a,B}(n) = n + C_B(a)$ for all $n \geq 1$. If B has at least two distinct prime divisors, there is no uniform bound on when the increments of $S_{a,B}(n)$ become permanently equal to one.*

Proof. Let $B = p^e$. The unit-component restriction forces $p \nmid a$. For odd p , the order d_p divides $p - 1$, which divides $p^e - 1 = B - 1$. Hence the order condition is satisfied from the start, and

$$S_{a,B}(n) = \left\lfloor \frac{ne + c_p}{e} \right\rfloor = n + \left\lfloor \frac{c_p}{e} \right\rfloor.$$

For $p = 2$ the binary formula gives the same conclusion.

Now suppose B has at least two prime divisors, and distinguish one of them, p_0 , with exponent e_0 in B . Choose any sufficiently large integer $C \geq 2$. By the Chinese remainder theorem choose $a \geq 2$ satisfying

$$\begin{aligned} a &\equiv p_0 \pmod{p_0^2}, \\ a &\equiv 1 + q^{e_q C} \pmod{q^{e_q C+1}} \quad \text{for each other prime } q \mid B. \end{aligned}$$

These moduli are pairwise coprime. The first condition gives $\nu_{p_0}(a) = 1$. Every other component is a unit congruent to 1 modulo q , with $c_q = e_q C$. At $q = 2$, the requirement $e_q C \geq 2$ ensures that $\nu_2(a + 1) = 1$, so the same equality for c_2 holds. There is no multiplicative-order obstruction. Formula (32) is therefore exact from $n = 1$ and becomes

$$S_{a,B}(n) = \min \left\{ n + C, \left\lfloor \frac{B^n}{e_0} \right\rfloor \right\}. \quad (33)$$

Fix any prescribed stage $n \geq 2$ and take $C \geq \lfloor B^n / e_0 \rfloor$. At both $n - 1$ and n the exponential term is the minimum. Their difference is

$$\left\lfloor \frac{B^n}{e_0} \right\rfloor - \left\lfloor \frac{B^{n-1}}{e_0} \right\rfloor > 1.$$

For completeness, $B \geq 6$ and $e_0 \leq B$, so the difference of the unrounded quantities is at least $B - 1 \geq 5$ for $n \geq 2$. Yet eventually the linear term wins. Since the prescribed stage can be arbitrarily late, there is no uniform onset bound. $\square$

This isolates the cause of the decimal counterexample: simultaneous components associated with different primes, some dividing the base and others not. Extremely fast growth by itself is not the obstruction.

12 What changes for genuine tetration?

Define the integer tower sequence by

$$T_0(a) = 1, \quad T_{h+1}(a) = a^{T_h(a)}.$$

For $h \geq 1$ this is $a \uparrow\uparrow h$. In contrast, $X_b(a) = a^{10^b}$ satisfies $X_{b+1}(a) = X_b(a)^{10}$. The two recurrences are fundamentally different.

Let $\Delta_h = T_{h+1}(a) - T_h(a)$. The tower recurrence yields

$$\Delta_h = a^{T_{h-1}(a)} (a^{\Delta_{h-1}} - 1) \quad (h \geq 2), \quad (34)$$

and for $h = 1$ the exponent in the second factor is $a - 1$. Local valuations are propagated through *previous tower differences*, not through the predetermined exponent $9 \cdot 10^b$.

12.1 A short derivation for base 3

The following recovers a known frozen-digit formula in the tetration literature [4]; it is included to show the mechanism directly.

Proposition 12.1. *For $T_h = T_h(3)$ and $h \geq 1$,*

$$\nu_2(T_{h+1} - T_h) = 2h + 1, \quad \nu_5(T_{h+1} - T_h) = h - 1.$$

Consequently

$$\nu_{10}(T_{h+1} - T_h) = h - 1.$$

Proof. The initial difference is $27 - 3 = 24$, with valuations 3 and 0. Every difference is divisible by 4. Since $C_2(3) = 2$ and $\nu_5(3^4 - 1) = 1$, lifting applied to (34) increases the binary valuation by 2 and the quinary valuation by 1 at each step. Induction gives both formulas. $\square$

Because the decimal valuations strictly increase, the first difference in a sum of later tower differences cannot be canceled at its first nonzero digit. Thus $T_h(3)$ shares exactly $h - 1$ trailing digits with every later tower. This is an example where enormous growth and a simple exact digit count coexist.

12.2 The counterexample base behaves differently in a true tower

Proposition 12.2. *Suppose a is an odd multiple of 5. Put $s = \nu_5(a)$, $c = C_2(a)$, and $A = \nu_2(a - 1)$. For $h \geq 1$,*

$$\nu_{10}(T_{h+1}(a) - T_h(a)) = \min\{sT_{h-1}(a), hc + A\}. \quad (35)$$

Proof. For $h = 1$, factor $a^a - a = a(a^{a-1} - 1)$. The quinary valuation is s , while binary lifting gives $c + A$. At every subsequent stage, the second factor in (34) is -1 modulo 5, so the quinary valuation is $sT_{h-1}(a)$. Every tower difference is even; binary lifting adds c to its binary valuation at each step. This gives $hc + A$ and proves the result. $\square$

For $a = a_\star$, we have $s = 1$ and $A = c = 99$, so the genuine tower profile is

$$\min\{T_{h-1}(a_\star), 99(h+1)\}.$$

At $h = 1$ it is 1, at $h = 2$ it is 297, and for every $h \geq 2$ it equals $99(h+1)$. To verify the last assertion, note $T_1(a_\star) > 297$ and $T_h(a_\star)$ increases at least geometrically, hence stays above the indicated linear bound. Its subsequent increments are 99, not 1. This does not conflict with the result about $a_\star^{10^b}$; it emphasizes why the two problems must be kept separate.

13 Exact verification and reproducibility

13.1 Certificates that do not expand the huge powers

If a formula predicts that a difference has radix- B valuation s , compute it modulo B^{s+1} . A nonzero residue equal to dB^s with $1 \leq d < B$ proves that its valuation is exactly s . Modular exponentiation performs this check without constructing the full integer powers.

For the primary example, the independent certificates are

$$a_\star^{10^3} - a_\star^{10^2} \equiv 4 \cdot 10^{100} \pmod{10^{101}}, \quad (36)$$

$$a_\star^{10^4} - a_\star^{10^3} \equiv 5 \cdot 10^{102} \pmod{10^{103}}. \quad (37)$$

They give $S_{a_\star}(2) = 100$ and $S_{a_\star}(3) = 102$ directly. The proofs earlier in the note do not depend on these calculations.

13.2 Included programs and actual test results

The archive contains `code/stabilization.py`, implementing the formulas, and `code/verify.py`, which generates the result files. Both use only the Python standard library and are written for Python 3.9 or later. The run packaged with this note used Python 3.13.5.

Check category	Completed checks
Primary-example modular certificates	7
Small decimal bases, $2 \leq a \leq 2000$	8995
Delayed-family modular certificates	64
Residue-class tests of the minimum table	100
Stage-minimum modular certificates	6
General-radix modular certificates, $2 \leq B \leq 35$	12489
Nonconsecutive decimal certificates	432
Period-two radix-14 certificates	4
Idempotent-limit tests	20

All completed checks passed. The general-radix sweep deliberately skipped 1111 cases whose predicted valuation exceeded a cap of 160, to avoid constructing unnecessarily large verification moduli. No conclusion about those skipped cases is drawn from the sweep; they are covered by the mathematical theorem. The counts and exact certificates are preserved in `data/verification.json` and `data/primary_certificates.csv`.

The table test runs through $99 \leq K \leq 198$, covering every residue class modulo 20 repeatedly, and independently enumerates $j2^K \pm 1$ for $1 \leq j \leq 4$. Minimality over *all* integers is established by Theorem 6.2, not by that finite test.

13.3 Reproduction commands and computational scope

From the archive's root directory, run

```
python code/verify.py
pdflatex -interaction=nonstopmode -halt-on-error delayed_digit_stabilization.tex
pdflatex -interaction=nonstopmode -halt-on-error delayed_digit_stabilization.tex
```

A PowerShell build script and a POSIX shell build script are also included. The PDF uses standard TeX Live packages; no external bibliography tool is required. The source contains its bibliography.

The modular verifier represents B^n as an ordinary integer, so it is not an evaluator for arbitrary unexpanded up-arrow expressions. The symbolic minimum specification avoids expanding 2^K when a stage is too large. Radix factorization in the utility code uses trial division and is intended for small radices, not for cryptographic-size arbitrary integers. These implementation limits do not restrict the proved formulas.

No proof-assistant formalization is included. The verification layer is exact integer arithmetic plus independently computed residues, rather than floating-point evidence or a formal kernel check.

14 Conclusions and the remaining boundary

The selected conjecture has a negative answer. Its smallest counterexample is a 31-digit integer with a compact description, $3 \cdot 2^{99} + 1$. The full explanation is the exact profile

$$S_a(b) = \min\{t10^b, b + C\}$$

for mixed decimal bases. The affine branch always wins eventually, but there is no uniform bound on when it does so.

The result can be sharpened in several independent directions. The onset is given by a single monotone integer inequality. The least exceptional base at stage b is explicit, with a coefficient and sign periodic in b modulo 20. The exceptional set has an exact positive density, which falls doubly exponentially with the stage. Arbitrary-radix analogues reveal both multiplicative-order obstructions and a distinction between prime power and mixed-prime radices. Fast-growing changes of index transform the rate exactly as in (27).

These statements settle a modest, precisely posed open question and provide a broader structure around it. They do not settle analytic interpolation questions for tetration, transcendence questions about towers, or comparison problems for arbitrary ordinal-indexed fast-growing hierarchies. Those are different targets. Within the selected target, however, the answer is complete: a counterexample, its minimality, an exact family of stage minima, and quantitative descriptions of both delay and rarity.

A A compact standalone answer to the original question

The following proof can be read without any other section.

The proposed threshold is false. Let $a = 3 \cdot 2^{99} + 1$. Then $a \equiv 15 \pmod{25}$, so $v_5(a) = 1$, while $v_2(a - 1) = 99$ and $v_2(a + 1) = 1$. For $b \geq 2$,

$$a^{10^{b+1}} - a^{10^b} = a^{10^b} (a^{9 \cdot 10^b} - 1).$$

Since $5 \mid a$, its quinary valuation is 10^b . Binary lifting gives its binary valuation as

$$99 + 1 + b - 1 = b + 99.$$

Therefore $S_a(b) = \min\{10^b, b + 99\}$, and

$$D_a(3) = \min\{1000, 102\} - \min\{100, 101\} = 2.$$

Furthermore, for $a = 5^K + 1$ one obtains $S_a(b) = \min\{10^b, b + K\}$. Choosing $K = 10^m - m + 1$ gives $D_a(m + 1) = 2$ for every $m \geq 2$. Thus no fixed replacement threshold works for all bases.

B A minimal executable certificate

This independent fragment uses no formulas from the companion module:

```
a = 3 * 2**99 + 1

checks = [(2, 100, 4), (3, 102, 5)]
for b, s, expected_digit in checks:
    modulus = 10**(s + 1)
    residue = (pow(a, 10**(b + 1), modulus)
               - pow(a, 10**b, modulus)) % modulus
    assert residue == expected_digit * 10**s

assert 102 - 100 == 2
print("Counterexample verified exactly.")
```

The module's formula routine and the independent modular routine test different descriptions of the same quantities. Agreement between them is useful for catching transcription and indexing errors, while the proofs establish the universal claims.

References

- [1] Marco Ripà. *Stabilization of one new trailing digit of a^{10^b} as b increases, for every a not divisible by 10*. Mathematics Stack Exchange, question 5103929, posted 24 October 2025. <https://math.stackexchange.com/q/5103929>. Public page checked for this note on 19 September 2026; the retrieved page displayed no answers.
- [2] Marco Ripà. The congruence speed formula. *Notes on Number Theory and Discrete Mathematics* **27**(4) (2021), 43–61. DOI: <https://doi.org/10.7546/nntdm.2021.27.4.43-61>. Preprint: <https://arxiv.org/abs/2208.02622>.
- [3] Marco Ripà and Luca Onnis. Number of stable digits of any integer tetration. *Notes on Number Theory and Discrete Mathematics* **28**(3) (2022), 441–457. DOI: <https://doi.org/10.7546/nntdm.2022.28.3.441-457>. Preprint: <https://arxiv.org/abs/2210.07956>.
- [4] Marco Ripà. Graham’s number stable digits: An exact solution. *Notes on Number Theory and Discrete Mathematics* **31**(3) (2025), 607–616. Preprint version 2, 12 September 2025: <https://arxiv.org/abs/2411.00015>.